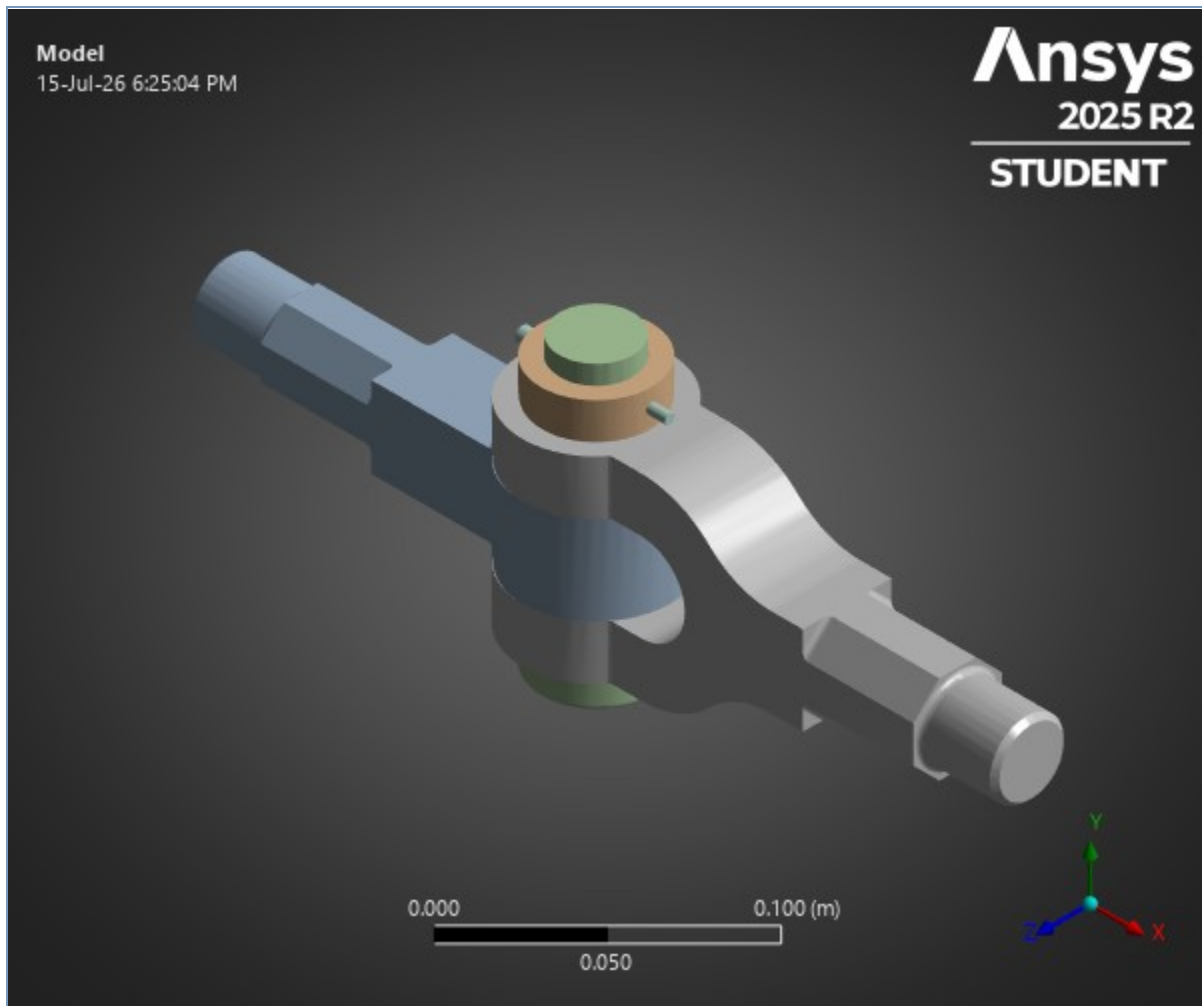




Project*

First Saved	Wednesday, July 15, 2026
Last Saved	Wednesday, July 15, 2026
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



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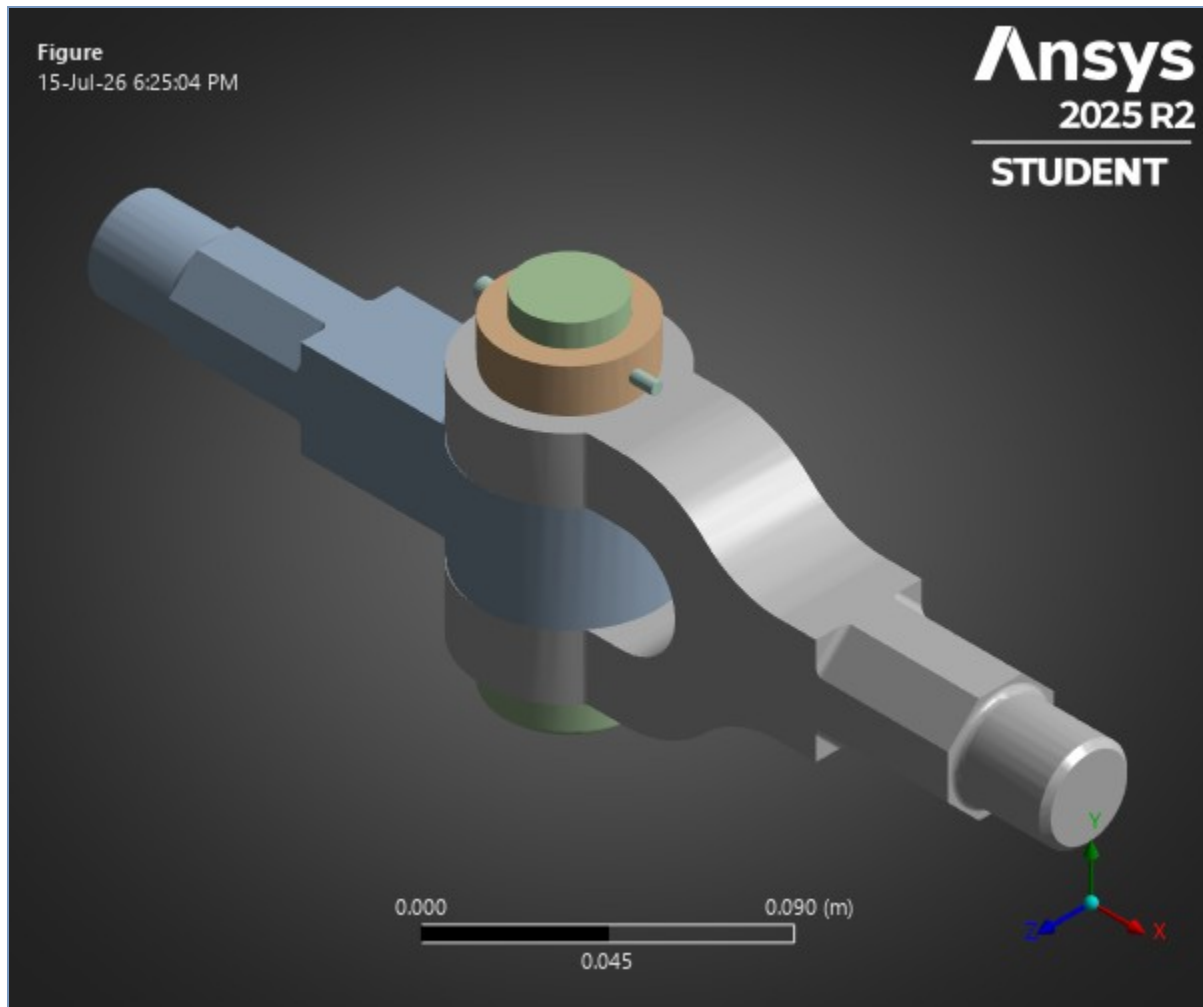
Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

FIGURE 1
Model (A4) > Figure

**TABLE 2****Model (A4) > Geometry Imports**

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3**Model (A4) > Geometry Imports > Geometry Import (A3)**

Object Name	<i>Geometry Import (A3)</i>
State	Solved
Definition	
Source	C:\Users\User\Downloads\knuckle joint\knuckle joint.stp
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes

Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	<i>Geometry</i>
State	Fully Defined
Definition	
Source	C:\Users\User\Downloads\knuckle joint\knuckle joint.stp
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	0.327 m
Length Y	0.117 m
Length Z	5.9998e-002 m
Properties	
Volume	5.838e-004 m ³
Mass	4.5829 kg
Scale Factor Value	1.
Statistics	
Bodies	5
Active Bodies	5
Nodes	11449
Elements	5532
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes

Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	<i>fork end:1 Solid1</i>	<i>eye end:1 Solid1</i>	<i>pin:1 Solid1</i>	<i>taper pin:1 Solid1</i>	<i>collor:1 Solid1</i>
State	Meshed				
Graphics Properties					
Visible	Yes				
Transparency	1				
Definition					
Suppressed	No				
Stiffness Behavior	Flexible				
Coordinate System	Default Coordinate System				
Reference Temperature	By Environment				
Treatment	None				
Material					
Assignment	Structural Steel				
Nonlinear Effects	Yes				
Thermal Strain Effects	Yes				
Bounding Box					
Length X	0.207 m	0.18 m	4.5e-002 m		6.e-002 m
Length Y	8.e-002 m	3.6e-002 m	0.117 m	1.5e-002 m	6.e-003 m
Length Z	5.9974e-002 m	5.9998e-002 m	4.5e-002 m		6.e-003 m
Properties					
Volume	2.8573e-004 m³	1.8896e-004 m³	9.4977e-005 m³	1.2945e-005 m³	1.1907e-006 m³
Mass	2.243 kg	1.4833 kg	0.74557 kg	0.10162 kg	9.3468e-003 kg
Centroid X	6.0087e-002 m	-5.0003e-002 m	1.3206e-005 m	1.4363e-004 m	-3.9666e-003 m
Centroid Y	-8.8412e-014 m	-1.0274e-013 m	-3.7591e-003 m	4.7e-002 m	
Centroid Z	8.5761e-009 m	5.8188e-010 m	-3.8048e-007 m	5.8629e-009 m	-7.5764e-017 m
Moment of Inertia Ip1	1.4406e-003 kg·m²	3.7113e-004 kg·m²	1.0137e-003 kg·m²	2.0946e-005 kg·m²	3.1074e-008 kg·m²
Moment of Inertia Ip2	6.9618e-003 kg·m²	3.9631e-003 kg·m²	1.095e-004 kg·m²	3.7155e-005 kg·m²	2.6994e-006 kg·m²
Moment of Inertia Ip3	7.6946e-003 kg·m²	3.8586e-003 kg·m²	1.0133e-003 kg·m²	2.0101e-005 kg·m²	2.6994e-006 kg·m²
Statistics					
Nodes	3149	3838	2448	928	1086
Elements	1567	2048	1285	434	198

Mesh Metric

None

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (A4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	6
Active Contacts	6
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0

Body Interactions	0
Active Body Interactions	0

TABLE 9
Model (A4) > Connections > Contacts

Object Name	<i>Contacts</i>
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	8.8111e-004 m
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75. °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	6
Active Connections	6

TABLE 10
Model (A4) > Connections > Contacts > Contact Regions

Object Name	Contact Region	Contact Region 2	Contact Region 3	Contact Region 4	Contact Region 5	Contact Region 6
State	Fully Defined					
Scope						
Scoping Method	Geometry Selection					
Contact	2 Faces	3 Faces	1 Face			2 Faces
Target	2 Faces	3 Faces	1 Face			
Contact Bodies	fork end:1 Solid1		eye end:1 Solid1	pin:1 Solid1		taper pin:1 Solid1
Target Bodies	eye end:1 Solid1	pin:1 Solid1		taper pin:1 Solid1	collor:1 Solid1	
Protected	No					
Definition						
Type	Bonded					
Scope Mode	Automatic					
Behavior	Program Controlled					
Trim Contact	Program Controlled					
Trim Tolerance	8.8111e-004 m					
Contact APDL Name						
Target APDL Name						

Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Program Controlled
Small Sliding	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

Mesh

TABLE 11
Model (A4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	0.35245 m
Average Surface Area	1.0302e-003 m ²
Minimum Edge Length	1.2807e-003 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272

Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	11449
Elements	5532
Show Detailed Statistics	No

Static Structural (A5)

TABLE 12
Model (A4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 13
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off

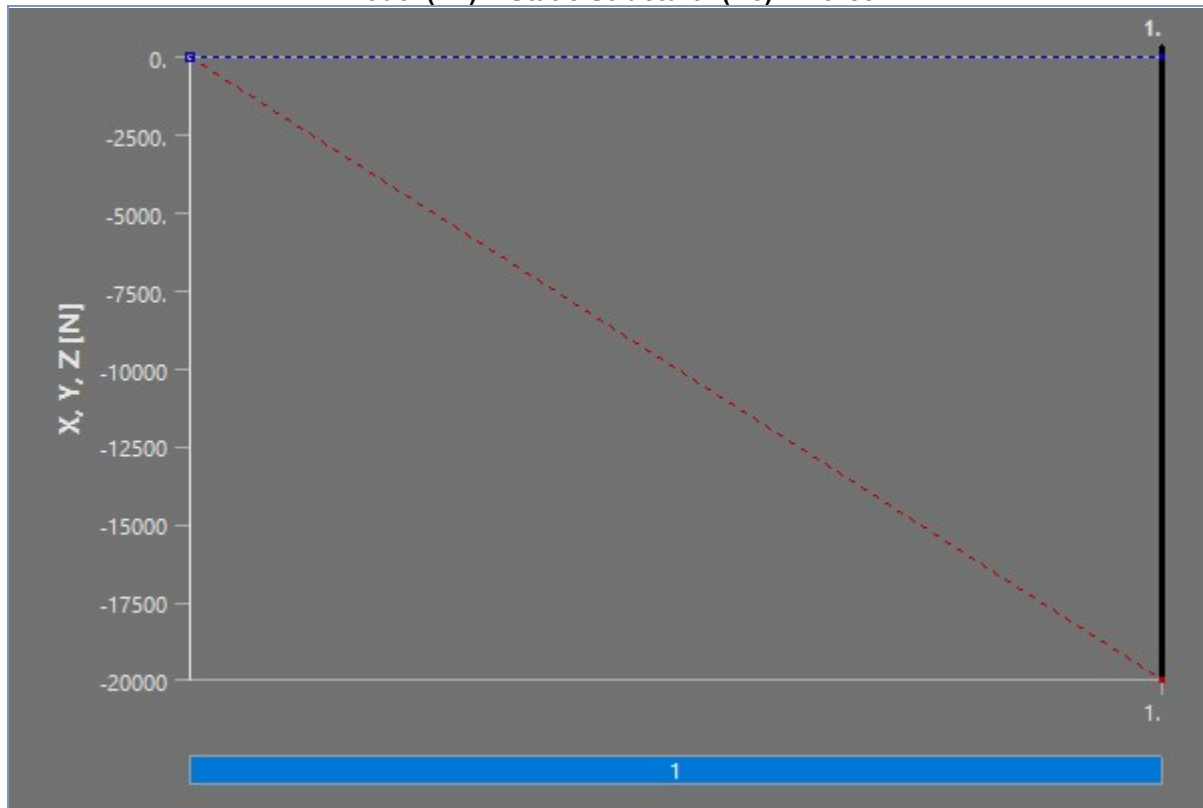
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\User\Downloads\knuckle joint\knuckle joint_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 14
Model (A4) > Static Structural (A5) > Loads

Object Name	Fixed Support	Force
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Face	
Definition		
Type	Fixed Support	Force
Suppressed	No	

Define By		Components
Applied By		Surface Effect
Coordinate System		Global Coordinate System
X Component		-20000 N (ramped)
Y Component		0. N (ramped)
Z Component		0. N (ramped)

FIGURE 2
Model (A4) > Static Structural (A5) > Force



Solution (A6)

TABLE 15
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	23. s
MAPDL Memory Used	259. MB
MAPDL Result File Size	4.25 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 16

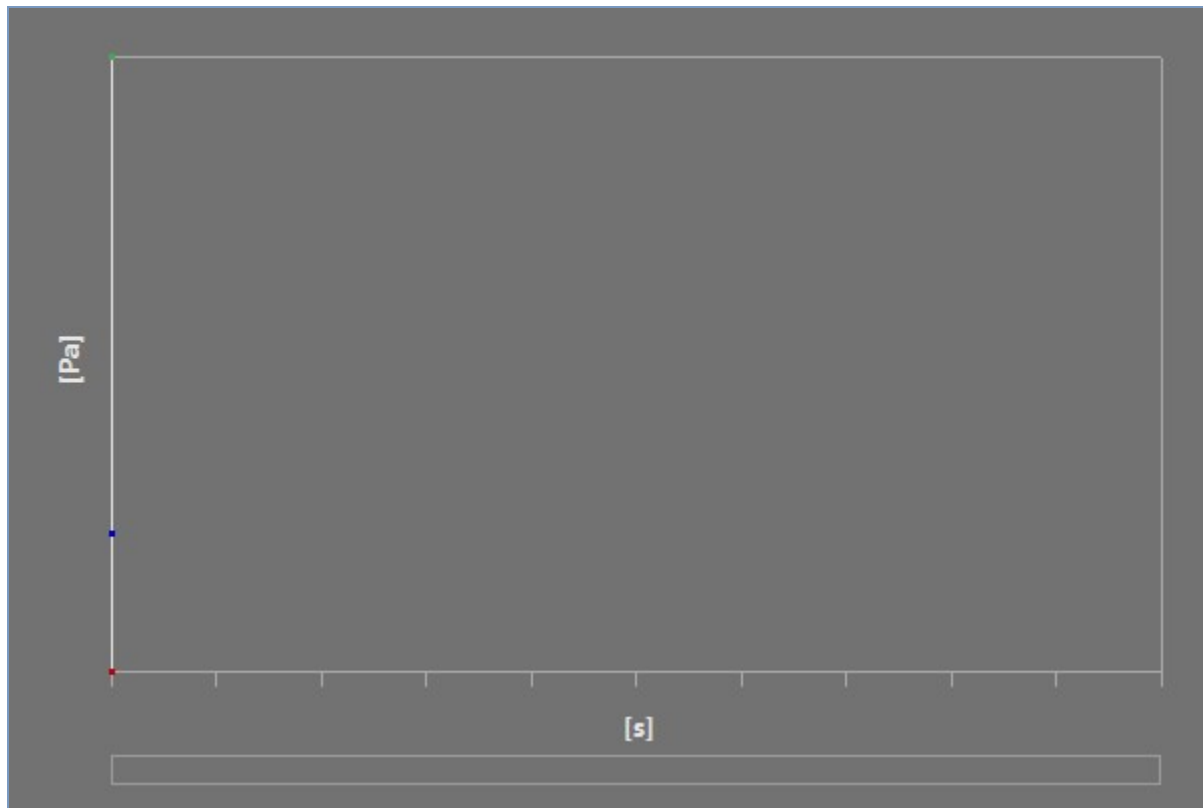
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 17**Model (A4) > Static Structural (A5) > Solution (A6) > Results**

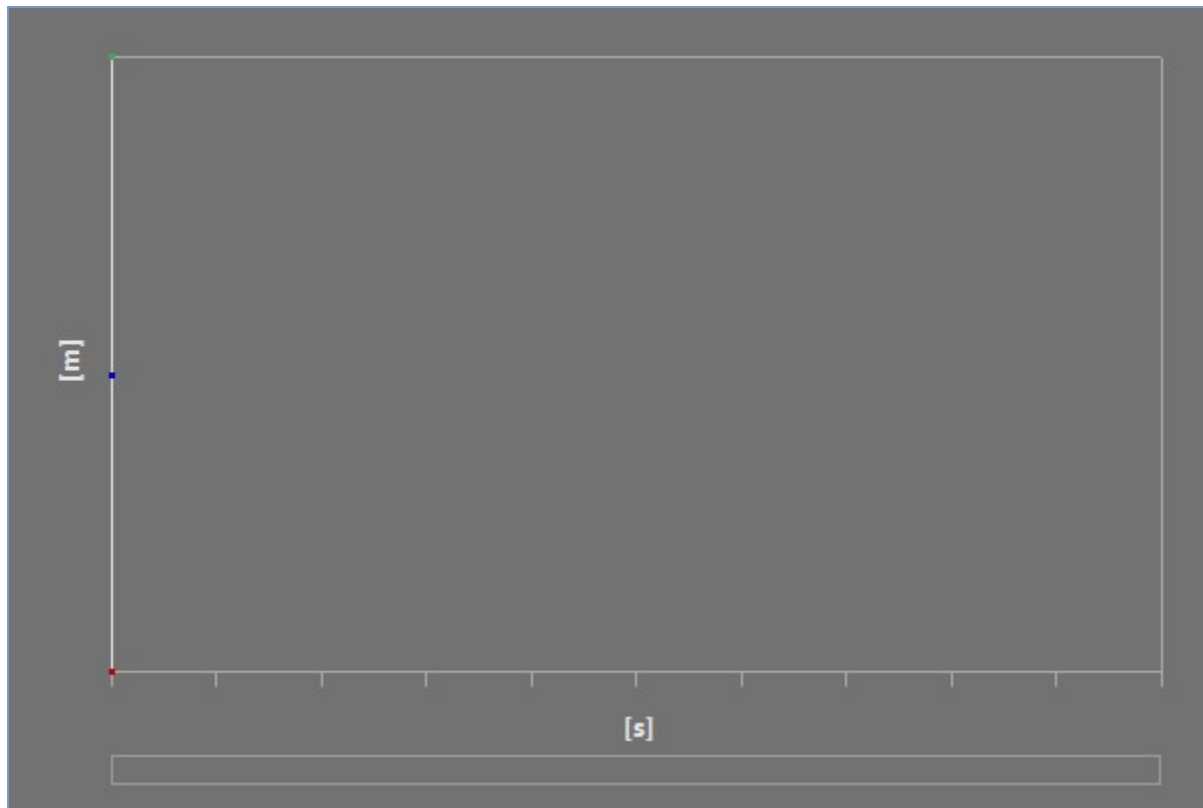
Object Name	Equivalent Stress	Directional Deformation
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Equivalent (von-Mises) Stress	Directional Deformation
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Orientation		X Axis
Coordinate System		Global Coordinate System
Integration Point Results		
Display Option	Averaged	
Average Across Bodies	No	
Results		
Minimum	5254.8 Pa	-3.2425e-005 m
Maximum	4.9453e+007 Pa	0. m
Average	1.1187e+007 Pa	-1.6776e-005 m
Minimum Occurs On	collor:1 Solid1	eye end:1 Solid1
Maximum Occurs On	fork end:1 Solid1	
Information		
Time	1. s	
Load Step	1	
Substep	1	
Iteration Number	1	

FIGURE 3**Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress**

**TABLE 18****Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress**

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	5254.8	4.9453e+007	1.1187e+007

FIGURE 4**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation**

**TABLE 19****Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation**

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.	-3.2425e-005	0.	-1.6776e-005

Material Data

Structural Steel

TABLE 20**Structural Steel > Constants**

Density	7850 kg m ⁻³
Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat	434 J kg ⁻¹ C ⁻¹
Thermal Conductivity	60.5 W m ⁻¹ C ⁻¹
Resistivity	1.7e-007 kg m ³ A ⁻² s ⁻³

TABLE 21**Structural Steel > Color**

Red	Green	Blue
132	139	179

TABLE 22**Structural Steel > Compressive Ultimate Strength**

Compressive Ultimate Strength Pa
0

TABLE 23

Structural Steel > Compressive Yield Strength

Compressive Yield Strength Pa
2.5e+008

TABLE 24**Structural Steel > Tensile Yield Strength**

Tensile Yield Strength Pa
2.5e+008

TABLE 25**Structural Steel > Tensile Ultimate Strength**

Tensile Ultimate Strength Pa
4.6e+008

TABLE 26**Structural Steel > Isotropic Secant Coefficient of Thermal Expansion**

Zero-Thermal-Strain Reference Temperature C
22

TABLE 27**Structural Steel > S-N Curve**

Alternating Stress Pa	Cycles	Mean Stress Pa
3.999e+009	10	0
2.827e+009	20	0
1.896e+009	50	0
1.413e+009	100	0
1.069e+009	200	0
4.41e+008	2000	0
2.62e+008	10000	0
2.14e+008	20000	0
1.38e+008	1.e+005	0
1.14e+008	2.e+005	0
8.62e+007	1.e+006	0

TABLE 28**Structural Steel > Strain-Life Parameters**

Strength Coefficient Pa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient Pa	Cyclic Strain Hardening Exponent
9.2e+008	-0.106	0.213	-0.47	1.e+009	0.2

TABLE 29**Structural Steel > Isotropic Elasticity**

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
2.e+011	0.3	1.6667e+011	7.6923e+010	

TABLE 30**Structural Steel > Isotropic Relative Permeability**

Relative Permeability
10000